

Analysis - High-Demand, Low_Supply Book Analysis

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Goal:

Let's analyze library dataset to find the books that have generated the most borrowings by number of copies available

Step 1: Load the dataset

We will use a dataset containing information about books:

catalog_position, title, author, year_published, genre, section, total_copies, times_borrowed, last_borrowed_date, rating, ratings_count, price, language, page_count, dimensions_width, dimension



```
In [7]: import pandas as pd
df = pd.read_csv("books.csv")
```

Step 2: Show some sample data

Show first 5 rows.

```
In [8]: df.head()
```

```
Out[8]:
```

	catalog_position	title	author	year_published	genre	section	total_co
0	A1-B1-0001	Enormous Room	E. E. Cummings	1949.0	Drama	Fiction	
1	A1-B1-0002	Plutarchi Vitae parallelae	Plutarch	1478.0	Biography & Autobiography	History	
2	A1-B1-0003	The water-babies	Charles Kingsley	1863.0	Fantasy	Children	
3	A1-B1-0004	Kipps	H. G. Wells	1900.0	Comedy	Fiction	
4	A1-B1-0005	Man and Superman	George Bernard Shaw	1903.0	Drama	Fiction	



Step 3: Calculate total copies / times borrowed ratio

We are going to create new column (borrowings_per_copy) with *times borrowed* to *total copies* ratio:

```
In [9]: df['borrowings_per_copy'] = df['times_borrowed'] / df['total_copies']
```

Step 4: Sort and print final results

We are going to sort the values, select first 50 and print the final results.

```
In [10]: sorted_df = df.sort_values(by='borrowings_per_copy', ascending=False)
top_50 = sorted_df.head(50)
top_50[['title', 'times_borrowed', 'total_copies', 'borrowings_per_copy']]
```

Out[10]:

	title	times_borrowed	total_copies	borrowings_per_copy
142	Where the Crawdads Sing	2256.0	5.0	451.200000
1010	The Handmaid's Tale	2248.0	5.0	449.600000
1032	Sapiens: A Brief History of Humankind	856.0	2.0	428.000000
1533	Atomic Habits: An Easy & Proven Way to Build G...	1193.0	3.0	397.666667
1051	Atomic Habits	1472.0	4.0	368.000000
231	Inferno	1471.0	4.0	367.750000
1428	The Hunger Games	1287.0	4.0	321.750000
553	The Subtle Art of Not Giving a F*ck	1844.0	6.0	307.333333
1718	The Alchemist	1530.0	5.0	306.000000
1312	Harry Potter and the Half-Blood Prince	1310.0	5.0	262.000000
393	When Breath Becomes Air	775.0	3.0	258.333333
212	Mockingjay	1201.0	5.0	240.200000
1839	Becoming	1419.0	6.0	236.500000
975	The Martian	1396.0	6.0	232.666667
742	Shantaram	454.0	2.0	227.000000
62	Catching Fire	1105.0	5.0	221.000000
1430	Fifty Shades of Grey	1083.0	5.0	216.600000
2026	Rich Dad, Poor Dad	864.0	4.0	216.000000
271	Fifty Shades Freed	826.0	4.0	206.500000
941	The Power of Habit: Why We Do What We Do in Li...	597.0	3.0	199.000000
970	Gone Girl	1165.0	6.0	194.166667
1784	The Book Thief	1128.0	6.0	188.000000
1276	Fifty Shades Darker	721.0	4.0	180.250000
1540	Harry Potter and the Order of the Phoenix	1061.0	6.0	176.833333
1086	Harry Potter and the Philosopher's Stone	1021.0	6.0	170.166667
396	To Kill a Mockingbird	848.0	5.0	169.600000
437	Sharp Objects	1137.0	7.0	162.428571

	title	times_borrowed	total_copies	borrowings_per_copy
1722	Tuesdays with Morrie	646.0	4.0	161.500000
131	A Game of Thrones	590.0	4.0	147.500000
1542	Big Little Lies	1178.0	8.0	147.250000
1089	The Stand	725.0	5.0	145.000000
869	Dune	715.0	5.0	143.000000
266	Harry Potter and the Goblet of Fire	846.0	6.0	141.000000
1052	The Fault in Our Stars	1059.0	8.0	132.375000
124	Being Mortal: Medicine and What Matters in the...	745.0	6.0	124.166667
16	A Thousand Splendid Suns	861.0	7.0	123.000000
1478	Twilight	600.0	5.0	120.000000
1079	Angels & Demons	353.0	3.0	117.666667
63	Divergent	328.0	3.0	109.333333
1933	Zero to One: Notes on Startups, or How to Build...	618.0	6.0	103.000000
1822	Und & Use Application Softwr	503.0	5.0	100.600000
766	On Writing: A Memoir of the Craft	197.0	2.0	98.500000
70	The Shining	480.0	5.0	96.000000
1592	The Very Hungry Caterpillar	556.0	6.0	92.666667
284	The Gunslinger	463.0	5.0	92.600000
1391	And the Mountains Echoed	457.0	5.0	91.400000
32	The Immortal Life of Henrietta Lacks	538.0	6.0	89.666667
65	Deception Point	354.0	4.0	88.500000
1200	Hatchet	425.0	5.0	85.000000
237	Fahrenheit 451	505.0	6.0	84.166667

Conclusion

- the results table shows books with the best times borrowed / total copies ratio
- the library should consider purchasing additional copies of the booksshowed in the table